

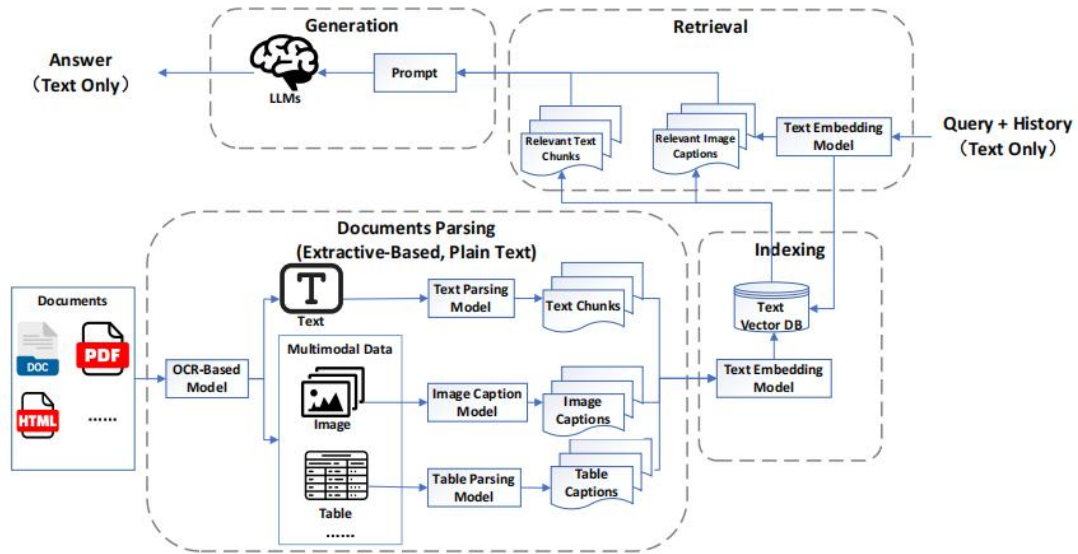


Fig. 1. The architecture of MRAG1.0 ([<Desc>1.png</Desc>](#)), often termed "pseudo-MRAG", closely resembles traditional RAG, consisting of three modules: Document Parsing and Indexing, Retrieval, and Generation. While the overall process remains largely unchanged, the key distinction lies in the Document Parsing stage. In this stage, specialized models are employed to convert diverse modal data into modality-specific captions. These captions are then stored alongside textual data for utilization in subsequent stages.

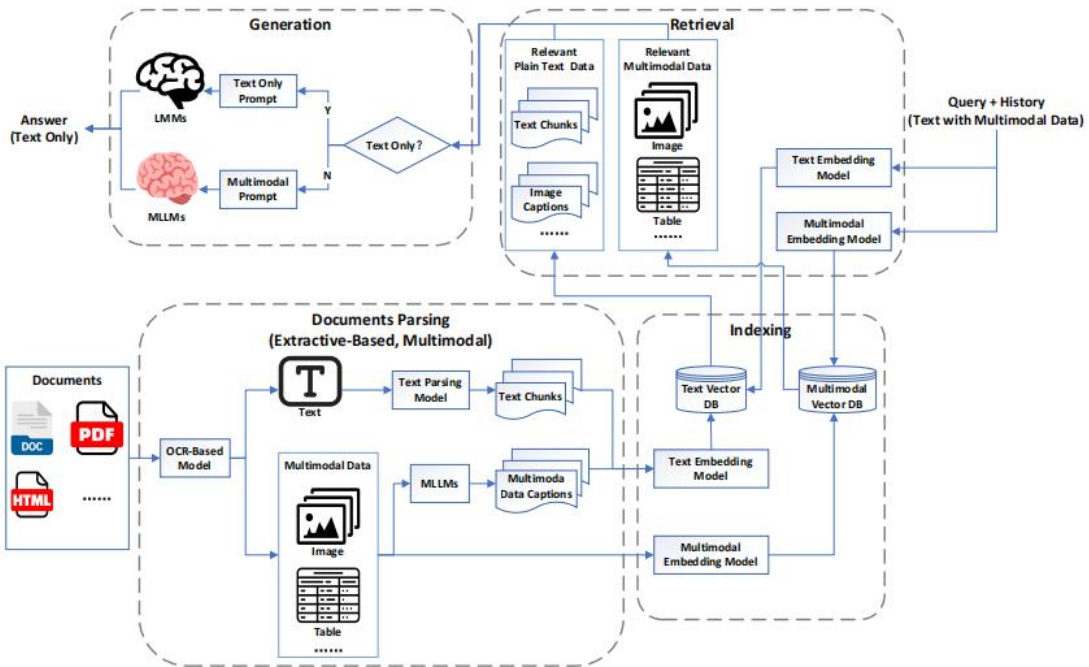


Fig. 2. The architecture of MRAG2.0 ([<Desc>2.png</Desc>](#)) retains multimodal data through document parsing and indexing, while introducing multimodal retrieval and MLLMs for answer generation, truly entering the multimodal era.



retrieval-augmented capabilities, the model can access relevant information beforehand, enabling the generation of precise and contextually accurate images.